Secure State Estimation for AUVs via T-N-L Observer with an Improved Adaptive Approach and Zonotopic Analysis

Chaojiang Liang, Yi Niu, Ben Niu, Xinjun Wang, Xiangyong Chen

***Abstract*—This paper addresses the problem of secure state estimation for autonomous underwater vehicles (AUVs) under attacks on measurement channels, unknown but bounded (UBB) disturbances, and measurement noise. First, a Takagi–Sugeno (T-S) fuzzy model is adopted to represent the nonlinear dynamics of AUVs. Then, an improved T-N-L observer with an adaptive mechanism is proposed, and the H_∞ technique is introduced to attenuate UBB disturbances and measurement noise. Subsequently, an encoding and decoding mechanism (EDM) is introduced within the T-S fuzzy framework to improve attack detectability. Furthermore, a zonotopic analysis framework is developed for attack reconstruction and isolation. Finally, simulations under sustained and intermittent attacks demonstrate the effectiveness of the proposed approach. In the main case, the detection rates on the two channels increase from 26.37% and 17.28% without EDM to 98.90% and 100.00% with EDM, respectively. Both detector configurations produce zero false alarms in these runs. The proposed method also reduces the root mean square of the estimation error norm by 31.40% relative to the adaptive observer without EDM.**

***Index Terms*—Autonomous underwater vehicles (AUVs), adaptive approach, H_∞ technique, Takagi-Sugeno (T-S) fuzzy system, zonotopic analysis.**

I. Introduction

Adaptive control has been studied for autonomous underwater vehicles (AUVs) under uncertain dynamics and external disturbances [1]. Attacks on measurement channels can degrade estimation and task execution, motivating secure state estimation for AUVs.

Attack detection methods include passive and proactive approaches. Passive methods leave transmitted data unchanged and include attack detection and isolation [2] and unknown input observer (UIO) attack reconstruction [3]. Adversaries with sufficient system knowledge can evade passive detection using stealthy attacks. This motivates proactive methods. Watermarking triggered by events was used to detect stealthy or replay attacks [4], [5]. In [6], a hybrid watermarking and moving target approach was proposed for systems with bounded noise. The work [7] combined dynamic watermarking with set membership estimation to maintain secure state estimation. Defenses using moving targets against adversaries with knowledge of the model have also been investigated [8]. Active detection against intelligent attacks was further studied in [9].

The problem of secure state estimation under sparse false data injection attacks has recently received considerable attention [10]–[12]. Among existing approaches, statistical and watermarking detectors mainly provide early warnings, whereas set membership and zonotopic methods characterize deterministic state and attack bounds under bounded uncertainty [13]–[15]. Set membership methods approximate feasible sets under bounded uncertainty [16]; zonotopes provide a compact generator representation. For example, in [17], a method based on zonotopes with encryption was proposed for resilient state estimation of nonlinear multirate networks under attacks and bounded uncertainties. The work [15] developed zonotopic observers to achieve secure estimation, reconstruction and isolation of attacks in cyber physical systems (CPSs) under bounded disturbances. Further studies appear in [18], [19]. Interval observers and AUV zonotopic methods have also been studied for bounded uncertainty and fault detection [20], [21]. Coding has also been used for distributed filtering under denial-of-service (DoS) attacks [22]. Distributed secure estimators exchange information over a graph and use sorting/filtering or adaptive saturation [23], [24]. In this paper, a centralized AUV observer isolates channels using zonotopic tests assisted by EDM. A distributed extension would require additional graph and communication assumptions. However, secure state estimation for AUVs has received limited attention compared with general networked CPSs. Practical AUV operation involves uncertain hydrodynamics, limited underwater communication, and restricted energy, which motivate the integrated design considered in this work [25], [26].

To model nonlinear AUV dynamics, a T-S fuzzy model

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with unmeasurable premise variables is employed [27]. T-S fuzzy methods have also been used for fault detection in unmanned surface vehicles and dynamic positioning of marine vehicles [28], [29]. Secure AUV estimation balances modeling accuracy, computational cost, and robustness to bounded uncertainty and attacks.

Motivated by [26], [30], we combine an improved adaptive T-N-L observer and an encoding and decoding mechanism (EDM) for secure AUV state estimation. The contributions are:

1) Compared with fault detection methods based on observers or zonotopes for AUV and T-S fuzzy systems [26], [31], an adaptive H_∞ T-N-L observer is developed for secure state estimation of AUVs under sparse sensor attacks, unknown but bounded disturbances, and measurement noise. The adaptive mechanism and H_∞ performance index jointly provide resilient estimation and disturbance attenuation under the stated admissible measurement masks.

2) An EDM based on the encoding and decoding principle in [30] is introduced to improve the detection of sparse sensor attacks. Compared with the attack detection methods in [3], [32], [33], the proposed method incorporates EDM into the AUV T-S fuzzy framework. It achieves higher attack detection rates than the corresponding detector without EDM under the same conditions.

3) A unified secure state estimation framework integrating the adaptive H_∞ T-N-L observer, attack detection using EDM, and zonotopic analysis is established. Compared with zonotopic secure estimation for general CPSs [15], this framework combines attack reconstruction, isolation, and state estimation for the reduced AUV T-S model.

Notation: $\mathbb{R}^n$ and $\mathbb{R}^{m\times n}$ denote real vectors and matrices. I_n, $\emptyset$, $(\cdot)^T$, and $(\cdot)^\dagger$ denote the identity, empty set, transpose, and pseudoinverse. $P \succ 0$ means positive definiteness; $*$ denotes a symmetric block. The Euclidean/spectral norm is $\|\cdot\|_2$, diag forms a diagonal or block diagonal matrix, and supp denotes support.

II. Preliminaries and System Model

A. Preliminaries

Definition 1 [34]: A box is defined as $[p,q] = \{x \in \mathbb{R}^n : p_i \le x_i \le q_i,\ i = 1,\dots,n\}$, where $p,q \in \mathbb{R}^n$ specify its lower and upper bounds, respectively.

Definition 2 [34]: A zonotope is defined as $\langle p, H\rangle = \{p + H\xi : \xi \in \mathbb{R}^m,\ \|\xi\|_\infty \le 1\}$, where $p \in \mathbb{R}^n$ and $H \in \mathbb{R}^{n\times m}$ are its center and generator matrix, respectively.

Definition 3 [14]: For a bounded set $\mathbf{G} \subset \mathbb{R}^n$, its interval hull is the smallest box containing $\mathbf{G}$, i.e.,

$$\mathbf{G} \subseteq \mathrm{Box}(\mathbf{G}) = [c,d], \tag{1}$$

where c and d denote the lower and upper bounds of the enclosing box, respectively.

Definition 4 [35]: The Minkowski sum of two sets $\mathbf{Y}$ and $\mathbf{Z}$ is defined as $\mathbf{Y} \oplus \mathbf{Z} = \{y + z : y \in \mathbf{Y},\ z \in \mathbf{Z}\}$, where $\oplus$ denotes the Minkowski sum operator.

Property 1 [36]: For a zonotope $\mathcal{Z} = \langle \bar{p}_k, \bar{H}_k\rangle$, the generators are retained if $m \le q$. Otherwise, the columns are ranked by the weighted norm $\|W\bar{h}_j\|_2$. Retaining the $q-n$ largest columns gives the partition $\bar{H}_k = [(\bar{H}_k)_> \ (\bar{H}_k)_<]$, where the discarded columns are enclosed by $\mathrm{rs}((\bar{H}_k)_<) = \mathrm{diag}(|(\bar{H}_k)_<|\mathbf{1})$. Then

$$\begin{aligned}\mathcal{Z} &\subseteq \left\langle \bar{p}_k,\ [(\bar{H}_k)_>\ \ \mathrm{rs}((\bar{H}_k)_<)]\right\rangle \\ &= \left\langle \bar{p}_k, \downarrow_{q,W} (\bar{H}_k)\right\rangle,\end{aligned} \tag{2}$$

where $q \ge n$ is the maximum number of generators and $W \succ 0$ is a weighting matrix; $W = I$ yields the standard reduction operator $\downarrow_q$. More generators can tighten the enclosure but increase computational cost. More aggressive reduction may widen residual bounds and reduce sensitivity to weak attacks.

B. Problem Formulation

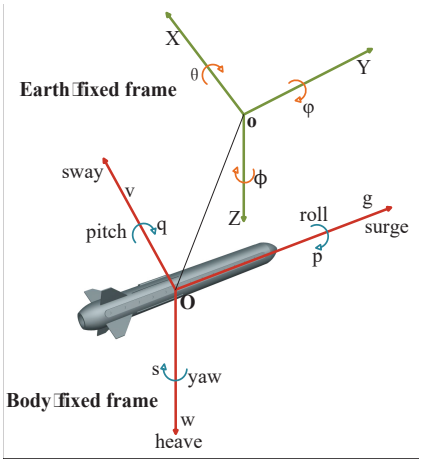


Fig. 1. Body and earth reference frames of AUVs.

Following the 6 DoF AUV model in [25] and the frame convention in Fig. 1, we consider the depth plane involving heave and pitch.

With $X_c = Y_c = Z_c = 0$, $W = F$, $\theta = p = 0$, and, at constant surge speed g_0, $v = s = 0$, the reduced dynamics are

$$(m + Z_{\dot{w}})\dot{w} + Z_w w + g_{m1} q = b_{g1} T_z + d_{g1}\varrho_z, \tag{3a}$$

$$(I_{yy} + M_{\dot{q}})\dot{q} - g_{m2} w + M_q q = b_{g2} M_{Ty} + d_{g2}\varrho_{Ty}, \tag{3b}$$

Here $g_{m1} = -mg_0 + X_{\dot{g}} g_0$ and $g_{m2} = mg_0 + X_{\dot{g}} g_0$; ϱ_z and ϱ_{Ty} collect unknown but bounded (UBB) modeling errors. UBB disturbances can represent uncertainty in hydrodynamic coefficients, linearization and reduction errors, nonlinear damping, buoyancy/mass changes, thruster uncertainty, and currents or waves. Zonotopic bounds represent these physical sources without a prescribed probability distribution.

The corresponding AUV kinematics are given by

$$\dot{Z} = w\cos\varphi - g_0\sin\varphi, \tag{4a}$$

$$\dot{\varphi} = q. \tag{4b}$$

For small φ, we use $\sin\varphi \approx \varphi$ and define $x = [w, q, Z, \varphi]^T$, $u = [T_z, M_{Ty}]^T$, and $\varrho = [\varrho_z, \varrho_{Ty}]^T$. Then

$$\dot{x}(t) = A_t x(t) + B_t u(t) + E_t \varrho(t), \tag{5}$$

where $A_t = -M^{-1}A_r$, $B_t = M^{-1}B_r$, and $E_t = M^{-1}E_r$; the kinematic rows give $\dot{Z} = w\cos\varphi - g_0\varphi$ and $\dot{\varphi} = q$. With

$$M = \begin{bmatrix} m + Z_{\dot{w}} & 0 & 0 & 0 \\ 0 & I_{yy} + M_{\dot{q}} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

$$A_r = \begin{bmatrix} Z_w & -mg_0 + X_{\dot{g}}g_0 & 0 & 0 \\ -mg_0 - X_{\dot{g}}g_0 & M_q & 0 & 0 \\ -\cos(\varphi(t)) & 0 & 0 & g_0 \\ 0 & -1 & 0 & 0 \end{bmatrix},$$

$$B_r = \begin{bmatrix} \mathrm{diag}(b_{g1}, b_{g2}) \\ 0_{2\times 2} \end{bmatrix}, \quad E_r = \begin{bmatrix} \mathrm{diag}(d_{g1}, d_{g2}) \\ 0_{2\times 2} \end{bmatrix}.$$

Zero order hold over one sampling interval gives

$$x_{k+1} = A_k x_k + B_k u_k + E_k \varrho_k. \tag{6}$$

Let $\iota_k = \cos\varphi_k \in [\iota_{\min}, \iota_{\max}]$. Since φ_k is not directly measured, the observer uses $\hat{\iota}_k = \cos\hat{\varphi}_k$. The mismatch is bounded by $d_k^p \in \mathcal{D}^p = \langle 0, H_p\rangle$ and included in the observer design.

Rule i: If ι_k is $\mathcal{G}_i(\iota_k)$, then

$$\begin{cases} x_{k+1} = A_i x_k + B_i u_k + E_i \varrho_k, \\ y_k = C x_k + D v_k. \end{cases} \tag{7}$$

where $i = 1, 2$. $\iota_k = \cos(\varphi_k)$ is the plant premise variable and $\mathcal{G}_i(\iota_k)$ is the membership function. The observer uses $\hat{\iota}_k = \cos(\hat{\varphi}_k)$, with $\hat{\sigma}_i(\hat{\iota}_k) = \mathcal{G}_i(\hat{\iota}_k)/\sum_{j=1}^{2}\mathcal{G}_j(\hat{\iota}_k)$. The variables $y_k \in \mathbb{R}^{n_y}$, $v_k \in \mathbb{R}^{n_v}$, $\varrho_k \in \mathbb{R}^{n_\varrho}$, and $d_k^p \in \mathbb{R}^{n_x}$ denote the measured output, measurement noise, disturbance, and bounded premise mismatch term, respectively. The matrices $C \in \mathbb{R}^{n_y\times n_x}$ and $D \in \mathbb{R}^{n_y\times n_v}$ are known constant matrices. For subsystem i, the discrete time matrices A_i, B_i, and E_i are obtained as

$$A_i = e^{A_{ti}\Delta t}, \quad B_i = \int_0^{\Delta t} e^{A_{ti}\chi} d\chi B_t,$$

$$E_i = \int_0^{\Delta t} e^{A_{ti}\chi} d\chi E_t$$

The continuous time matrix $A_t(\iota)$ is affine in the premise variable through the kinematic row. We use the endpoint matrices A_{s1} and A_{s2} to construct the sampled data vertices $A_i = e^{A_{ti}\Delta t}$ and the corresponding input matrices from zero order hold B_i and E_i. The resulting convex T-S model approximates the sampled dynamics. The mismatch due to $\hat{\iota}_k$ is included in the bounded modeling term. The continuous matrices are

$$A_{ti} = -M^{-1}A_{si},$$

$$A_{s1} = \begin{bmatrix} Z_w & -mg_0 + X_{\dot{g}}g_0 & 0 & 0 \\ -mg_0 - X_{\dot{g}}g_0 & M_q & 0 & 0 \\ -\iota_{\max} & 0 & 0 & g_0 \\ 0 & -1 & 0 & 0 \end{bmatrix},$$

$$A_{s2} = \begin{bmatrix} Z_w & -mg_0 + X_{\dot{g}}g_0 & 0 & 0 \\ -mg_0 - X_{\dot{g}}g_0 & M_q & 0 & 0 \\ -\iota_{\min} & 0 & 0 & g_0 \\ 0 & -1 & 0 & 0 \end{bmatrix},$$

$$h_1(\iota_k)\iota_{\max} + h_2(\iota_k)\iota_{\min} = \iota_k, \quad h_1(\iota_k) + h_2(\iota_k) = 1.$$

Then, the T-S fuzzy rules can be transformed into

$$\begin{cases} x_{k+1} = A(\hat{\sigma}_k)x_k + B(\hat{\sigma}_k)u_k + E(\hat{\sigma}_k)\varrho_k + d_k^p, \\ y_k = Cx_k + Dv_k, \end{cases} \tag{8}$$

where $A(\hat{\sigma}_k) = \sum_{i=1}^{2}\hat{\sigma}_i(\hat{\iota}_k)A_i$, $B(\hat{\sigma}_k) = \sum_{i=1}^{2}\hat{\sigma}_i(\hat{\iota}_k)B_i$, and $E(\hat{\sigma}_k) = \sum_{i=1}^{2}\hat{\sigma}_i(\hat{\iota}_k)E_i$. The estimated weights satisfy $0 \le \hat{\sigma}_i(\hat{\iota}_k) \le 1$ and $\sum_{i=1}^{2}\hat{\sigma}_i(\hat{\iota}_k) = 1$. The term d_k^p belongs to the prescribed premise mismatch set $\mathcal{D}^p = \langle 0, H_p\rangle$.

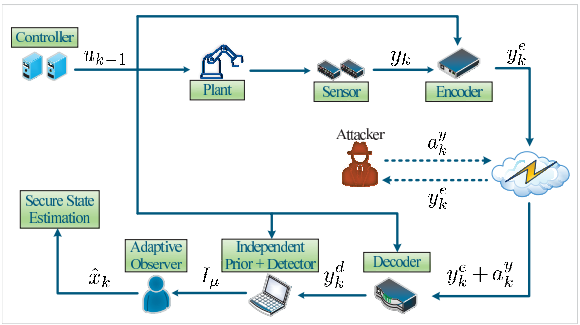


Fig. 2. Encoding, channel injection, decoding, and attack isolation using an independent prior.

Definition 5 [30]: The measurement is encoded before transmission according to

$$y_k^e = \alpha(u_{k-1}, y_k) = e^{-\beta_k\|u_{k-1}\|_2^2} \cdot z_k \cdot y_k, \tag{9}$$

where $y_k^e \in \mathbb{R}^{n_y}$ denotes the encoded measurement transmitted through an insecure network channel. $\beta_k > 0$ is the encoding intensity parameter, which varies with time. $z_k = \mathrm{diag}(\kappa_k^1, \kappa_k^2, \ldots, \kappa_k^{n_y})$ is an invertible diagonal encoding matrix that varies with time. $\|u_{k-1}\|_2^2$ is the squared Euclidean norm of the previous control input.

Threat model: The attacker can modify only the encoded packet between the trusted encoder and decoder. Attacks before encoding, replay, delay, and packet loss are not considered.

Remark 1: The bounded parameters β_k and z_k, whose diagonal entries are nonzero, are synchronized by a shared pseudorandom number generator (PRNG) sequence [9]. EDM assumes a trusted encoder, decoder, and seed; it does not provide cryptographic security. The bounded parameter β_k determines scaling that depends on the input, and z_k determines scaling for each channel; its diagonal entries are bounded away from zero. Their ranges are chosen according to admissible inputs, sensor/packet

resolution, and detection sensitivity. Exact synchronized decoding preserves noise before encoding. However, $G_k = e^{\beta_k \|u_{k-1}\|_2^2} z_k^{-1}$ can amplify errors after encoding, quantization, and synchronization mismatch, which limits the usable gains.

At the receiver, the encoded measurement is decoded as

$$y_k^d = \alpha^{-1}(y_k^e) = e^{\beta_k \|u_{k-1}\|_2^2} \cdot z_k^{-1} \cdot y_k^e. \tag{10}$$

In the absence of attacks, the decoded measurement satisfies

$$y_k^d = e^{\beta_k \|u_{k-1}\|_2^2} \cdot z_k^{-1} \cdot (e^{-\beta_k \|u_{k-1}\|_2^2} \cdot z_k) \cdot y_k = y_k, \tag{11}$$

Thus, without attacks, EDM leaves the measurement and observer update unchanged.

The notation F_k is equivalent to z_k. The decoder receives $y_k^r = y_k^e + a_k^y$. The decoded signal supplied to the observer is then

$$\begin{aligned} y_k^d &= \alpha^{-1}(y_k^e + a_k^y) \\ &= e^{\beta_k \|u_{k-1}\|_2^2} \cdot z_k^{-1} \cdot (y_k^e + a_k^y) \\ &= y_k + \underbrace{e^{\beta_k \|u_{k-1}\|_2^2} \cdot z_k^{-1} \cdot a_k^y}_{\Delta y_k}. \end{aligned} \tag{12}$$

We define $\Delta y_k = e^{\beta_k \|u_{k-1}\|_2^2} z_k^{-1} a_k^y$, so the attack on the encoded packet is scaled by the gain that varies with time G_k. Hence,

$$\begin{cases} x_{k+1} = A(\hat{\sigma}_k)x_k + B(\hat{\sigma}_k)u_k + E(\hat{\sigma}_k)\varrho_k + d_k^p, \\ y_k^d = Cx_k + Dv_k + \Delta y_k, \end{cases} \tag{13}$$

Let $\mathcal{U} \subseteq \{1, \ldots, n_y\}$ be the attackable channels and $\mathcal{S}_s = \{S \subseteq \mathcal{U} : |S| \le s\}$. Define $I_S = \mathrm{diag}(\mathbf{1}_{j \notin S})$ and $C_S = I_S C$. Observer matrices are designed for every admissible S and fuzzy rule. The common P result applies to segments with fixed masks; switching requires transition conditions.

Assumption 1: The initial state x_0, measurement noise v_k, and disturbance ϱ_k belong to known zonotopes:

$$x_0 \in \langle p_0, H_0 \rangle, \;\; v_k \in \mathbf{V} = \langle 0, H_v \rangle, \;\; \varrho_k \in \mathbf{W} = \langle 0, H_w \rangle,$$

where p_0, H_0, H_v and H_w are a known vector and known matrices, respectively.

Remark 2: The zonotopes define admissible uncertainty sets for deterministic reachability and can be estimated from specifications, calibration, repeated tests, and operating ranges. Safety margins are needed because the guarantees may fail when the actual uncertainty leaves these sets [20], [21], [26].

Assumption 2: For every fuzzy rule i and admissible set $S \in \mathcal{S}_s$, (A_i, C_S) is detectable.

Assumption 3: The attack support is contained in $\mathcal{U}$ and has sparsity at most s, $|\mathrm{supp}(a_k^y)| \le s \le s_{\max} < n_y$. Every attacked set satisfies Assumption 2.

III. Main results

A. Design of T-N-L observer with adaptive mechanism

In this subsection, we design an adaptive H_∞ T-N-L observer based on [15], [27], [37] as follows:

$$\begin{cases} \hat{x}_k = T_{S_k} A(\hat{\sigma}_{k-1})\hat{x}_{k-1} + T_{S_k} B(\hat{\sigma}_{k-1})u_{k-1} \\ \qquad + L_{S_k}(\hat{\sigma}_{k-1}) I_{S_{k-1}} (y_{k-1}^d - C\hat{x}_{k-1}) \\ \qquad + N_{S_k} I_{S_k} y_k^d, \\ \varepsilon_k = I_{S_k} y_k^d - I_{S_k} C \hat{x}_k, \end{cases} \tag{14}$$

where $\hat{x}_k$ is the posterior state estimate and ε_k denotes the posterior residual signal. The matrix I_{S_k} is selected after the current decoded measurement is tested, while $I_{S_{k-1}}$ is retained for the term using the previous measurement. The observer gain is scheduled by the estimated premise variable as $L_S(\hat{\sigma}_k) = \sum_{i=1}^{2} \hat{\sigma}_i(\hat{\imath}_k) L_{i,S} \in \mathbb{R}^{n_x \times n_y}$. For every admissible mask S, the matrices $T_S \in \mathbb{R}^{n_x \times n_x}$ and $N_S \in \mathbb{R}^{n_x \times n_y}$ are designed to satisfy

$$T_S + N_S I_S C = I_{n_x}. \tag{15}$$

Remark 3: The T-N-L structure adds T_S, N_S to the innovation gain L_S, with $T_S + N_S I_S C = I$; $T_S = I, N_S = 0$ recovers the Luenberger observer. This structure and its design freedom follow [15], [27], [37]. Here, "improved" denotes mask-dependent $T_S, N_S, L_{i,S}$, online selection through zonotopic adaptive channel isolation, and integration with H_∞ attenuation conditions. This observer, EDM, and zonotopic reconstruction form the complete framework. The result applies to segments with fixed masks; transitions require further analysis.

Below, S is omitted when the fixed mask is clear.

The estimation error e_k is defined as $e_k = x_k - \hat{x}_k$. For a fixed observer mode S, define $\tilde{\varrho}_k = [\varrho_k^T \;\; (d_k^p)^T]^T$, $n_{\tilde{\varrho}} = n_\varrho + n_x$, $\tilde{E}_i = [E_i \; I_{n_x}]$, $\tilde{E}(\hat{\sigma}_k) = [E(\hat{\sigma}_k) \; I_{n_x}]$, and $\tilde{H}_\varrho = \mathrm{diag}(H_w, H_p)$. The bounded term d_k^p represents the premise mismatch separately from ϱ_k. The error system can then be written as

$$\begin{aligned} e_k &= A_c(\hat{\sigma}_{k-1}, S)e_{k-1} \\ &\quad + B_c(\hat{\sigma}_{k-1}, S)f_k + C_c(\hat{\sigma}_{k-1}, S)a_k, \end{aligned} \tag{16}$$

where

$$\begin{aligned} A_c(\hat{\sigma}_{k-1}, S) &= T_S A_{k-1} - L_{k-1,S} I_S C, \\ B_c(\hat{\sigma}_{k-1}, S) &= [T_S \tilde{E}_{k-1} \quad - L_{k-1,S} I_S D \quad - N_S I_S D], \\ C_c(\hat{\sigma}_{k-1}, S) &= [-L_{k-1,S} I_S \quad - N_S I_S], \\ f_k &= [\tilde{\varrho}_{k-1}^T \quad v_{k-1}^T \quad v_k^T]^T, \\ a_k &= [\Delta y_{k-1}{}^T \quad \Delta y_k^T]^T. \end{aligned}$$

Remark 4: Write $A_{k-1} = A(\hat{\sigma}_{k-1})$, $\tilde{E}_{k-1} = \tilde{E}(\hat{\sigma}_{k-1})$, and $L_{k-1,S} = L_S(\hat{\sigma}_{k-1})$. Here, $f_k = [\tilde{\varrho}_{k-1}^T \; v_{k-1}^T \; v_k^T]^T$ collects the augmented disturbance, including the separately bounded premise mismatch term, together with the measurement noise terms at the previous and current sampling instants. Thus, f_k represents the exogenous uncertainty input in the H_∞ performance inequality. Correct isolation removes a_k on a segment with a fixed mask.

The following theorem gives the robust H_∞ observer design for each fuzzy rule and admissible mask.

Theorem 1: Fix a scalar $\gamma > 0$ and an admissible mask $S \in \mathcal{S}_s$. If there exist a positive definite matrix P and matrices $W_{i,S}$ and Y_S satisfying the LMI for every fuzzy rule i, then the corresponding observer matrices T_S, N_S, and $L_{i,S}$ are obtained from the solutions. A common P may be imposed across masks. This provides a bound for each design with a fixed mask but does not establish stability under arbitrary switching.

$$\begin{bmatrix} I_{n_x}-P & * & * & * & * \\ 0 & -\gamma^2 I_{n_{\tilde{e}}} & * & * & * \\ 0 & 0 & -\gamma^2 I_{n_v} & * & * \\ 0 & 0 & 0 & -\gamma^2 I_{n_v} & * \\ \xi_1 & \xi_2 & \xi_3 & \xi_4 & -P \end{bmatrix} \prec 0, \tag{17}$$

where

$$\begin{aligned} \xi_1 &= P\Theta_S^\dagger \Xi_1 A_i + Y_S \Psi_S \Xi_1 A_i - W_{i,S} I_S C, \\ \xi_2 &= P\Theta_S^\dagger \Xi_1 \tilde{E}_i + Y_S \Psi_S \Xi_1 \tilde{E}_i, \\ \xi_3 &= -W_{i,S} I_S D, \\ \xi_4 &= -P\Theta_S^\dagger \Xi_2 I_S D - Y_S \Psi_S \Xi_2 I_S D, \\ \Theta_S &= \begin{bmatrix} I_{n_x} \\ I_S C \end{bmatrix}, \Xi_1 = \begin{bmatrix} I_{n_x} \\ \mathbf{0} \end{bmatrix}, \Xi_2 = \begin{bmatrix} \mathbf{0} \\ I_{n_y} \end{bmatrix}, \\ \Psi_S &= I_{n_x+n_y} - \Theta_S \Theta_S^\dagger, \end{aligned}$$

and T_S, N_S and $L_{i,S}$ can be obtained by

$$\begin{aligned} T_S &= \Theta_S^\dagger \Xi_1 + P^{-1} Y_S \Psi_S \Xi_1, \\ N_S &= \Theta_S^\dagger \Xi_2 + P^{-1} Y_S \Psi_S \Xi_2, \\ L_{i,S} &= P^{-1} W_{i,S}, \end{aligned}$$

where $\Theta_S = [I_{n_x}^T \ (I_S C)^T]^T$ and $\Psi_S = I - \Theta_S \Theta_S^\dagger$. If the mask is correctly identified, the error system is stable within the corresponding segment with a fixed mask and satisfies

$$\sum_{k=k_0}^{N-1} e_k^T e_k \le \gamma^2 \sum_{k=k_0+1}^{N} f_k^T f_k + V(k_0), \tag{18}$$

where k_0 is the start of the segment with a fixed mask. For $e_{k_0} = 0$, $f \in \ell_2$, and $N \to \infty$, this gives $\|e\|_{\ell_2} \le \gamma \|f\|_{\ell_2}$, conditional on correct channel isolation.

Proof: Let $V(k) = e_k^T P e_k$. Substitution of the error system into $\Delta V(k) + e_{k-1}^T e_{k-1} - \gamma^2 f_k^T f_k \le 0$ and the Schur complement yield the LMI. Convexity extends the vertex conditions to the scheduled weights. Summing this dissipation inequality gives the bound in Theorem 1. The result applies within a segment with a fixed mask after correct isolation; it does not cover mask transitions. □

Remark 5: The induced ℓ_2 bound of the T-N-L observer is minimized by solving

$$\min \ \gamma^2, \quad \text{s.t. (17).} \tag{19}$$

The LMI solution gives $S_{0,S} = P^{-1} Y_S$. A common P may be used across masks; switching requires transition conditions.

B. Adaptive mechanism via zonotopic analysis

The state enclosure $\mathcal{X}_k^- = \langle c_k^-, H_k^- \rangle$ is computed without potentially attacked measurements. With $p_k^- = c_k^- - \hat{x}_k^-$, the corresponding set of prediction errors is obtained. Before using y_k^d, we form

$$\begin{aligned} \hat{x}_k^- &= A(\hat{\sigma}_{k-1})\hat{x}_{k-1} + B(\hat{\sigma}_{k-1})u_{k-1}, \\ e_k^- &= x_k - \hat{x}_k^-, \\ \mathcal{E}_k^- &= \langle p_k^-, H_k^- \rangle. \end{aligned} \tag{20}$$

The state enclosure is propagated using $A(\hat{\sigma}), B(\hat{\sigma})$ and generators $[A(\hat{\sigma})H^+, E(\hat{\sigma})H_w, H_p]$, followed by reduction that preserves enclosure. Trusted measurements can contract this set. Zero inclusion alone, however, does not establish that a measurement is free of attacks.

Theorem 2: If the T-S state is enclosed by a valid prior before using the current measurement, the amplified channel attack belongs to

$$\Delta y_k \in \Delta \widehat{y}_k = \langle p_k^{a_y}, H_k^{a_y} \rangle, \tag{21}$$

with

$$\begin{aligned} p_k^{a_y} &= y_k^d - C(p_k^- + \hat{x}_k^-), \\ H_k^{a_y} &= [CH_k^- \quad DH_v]. \end{aligned}$$

Proof: From $y_k^d = C(e_k^- + \hat{x}_k^-) + Dv_k + \Delta y_k$ and $e_k^- \in \langle p_k^-, H_k^- \rangle$, we obtain the center and generators stated above. Hence $\Delta \widehat{y}_k$ encloses the amplified attack under the assumed noise bounds. For $\mathcal{A}_{k,i} = \mathrm{Box}(\Delta \widehat{y}_{k,i})$, inclusion of zero indicates consistency with the model without attacks. If zero is excluded, channel i is declared attacked:

$$\begin{cases} \tau_{k,i}^y = 0, & 0 \in \mathcal{A}_{k,i} \quad \text{(no detection)}; \\ \tau_{k,i}^y = 1, & 0 \notin \mathcal{A}_{k,i} \quad \text{(attack detected)}. \end{cases}$$

□

The prior set is initialized from an admissible set of initial states and bounded uncertainties, with $p_k^- = c_k^- - \hat{x}_k^-$. Theorem 2 does not require a permanently trusted depth channel; this configuration is used in the Section IV simulation.

Remark 6: Let $G_k = e^{\beta_k \|u_{k-1}\|_2^2} \Gamma_k^{-1}$ and $r_{k,i} = \|e_i^T H_k^{a_y}\|_1$. Since the reconstructed attack center satisfies $p_{k,i}^{a_y} = G_{k,ii} a_{k,i}^y + \eta_{k,i}$ with $|\eta_{k,i}| \le r_{k,i}$, zero exclusion is guaranteed if

$$|a_{k,i}^y| > \frac{2r_{k,i}}{|G_{k,ii}|}.$$

Without EDM, $G_k = I$ and the sufficient threshold becomes $2r_{k,i}$. Thus, when $|G_{k,ii}| > 1$, EDM reduces the sufficient detection threshold. This deterministic comparison uses the same uncertainty bounds and does not establish superiority over all existing detectors.

Based on these detection flags, the adaptive mechanism is defined as

$$I_{S_k} = \mathrm{diag}\big(1 - \tau_{k,1}^y, 1 - \tau_{k,2}^y, \ldots, 1 - \tau_{k,n_y}^y\big), \tag{22}$$

where $\tau_{k,i}^y = 1$ when the ith measurement/communication channel is declared attacked; otherwise, $\tau_{k,i}^y = 0$.

Algorithm 1 summarizes the implementation. The flag $\tau^y_{k,i} = 1$ indicates a detected attack on channel i. The unknown attack a^y_k is not required as an input.

Algorithm 1 Implementation of secure state estimation

1: **Input:** measurements, controls, admissible initial set, trusted measurement update rule, and precomputed observer matrices.
2: **for** $k = 1$ to N **do**
3: Generate synchronized β_k, z_k and decode y^r_k to y^d_k.
4: Predict $\hat{x}^-_k$ and the independent state enclosure $\mathcal{X}^-_k$; set $p^-_k = c^-_k - \hat{x}^-_k$.
5: Reconstruct $\Delta\widehat{y}_k$; declare channel i attacked if $0 \notin \mathrm{Box}(\Delta\widehat{y}_{k,i})$.
6: Form I_{S_k} and select the precomputed matrices $T_{S_k}, N_{S_k}, L_{i,S_k}$.
7: Update $\hat{x}_k$ using the two measurement masks; contract $\mathcal{X}^-_k$ only with trusted measurements.
8: **end for**
9: **Output:** $I_{S_k}, \hat{x}_k$

IV. Simulation study

The AUV parameters follow [26]: $M = \mathrm{diag}(249.5328, 4932.254, 1, 1)$, $Z_w = 6.388$, $M_q = 1.174$, $X_{\dot{g}} = 25.522$, and $g_0 = 2$ in SI units. The input and disturbance gains are set to one. For $\Delta t = 0.02$ s and $\varphi_k \in [-\pi/6, \pi/6]$, the vertices obtained by zero order hold are given below for $i = 1, 2$. Values are rounded; the code uses full precision.

$$A_i \approx \begin{bmatrix} 0.999515 & 0.028603 & 0 & 0 \\ 0.001861 & 1.000022 & 0 & 0 \\ a_i & b_i & 1 & -0.04 \\ 0.000019 & 0.020000 & 0 & 1 \end{bmatrix},$$

$$B_i = E_i \approx 10^{-6} \begin{bmatrix} 80.129980 & 0.057996 \\ 0.074583 & 4.054968 \\ c_i & d_i \\ 0.000497 & 0.040550 \end{bmatrix}.$$

The entries that differ between vertices are

i	a_i	b_i	c_i	d_i
1	0.019995	−0.000114	0.801360	−0.000154
2	0.017316	−0.000152	0.693997	−0.000206

The observability rank using only velocity measurements is two, so a trusted depth measurement is included. Accordingly,

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \qquad D = I_3.$$

The attackable channels are $\mathcal{U} = \{1, 2\}$, with $s = 1$. The masks I_3, $\mathrm{diag}(0, 1, 1)$, and $\mathrm{diag}(1, 0, 1)$ all give observability rank four at both vertices. The initial conditions and generators are $x_1 = \hat{x}_1 = [0, 0, 2, 0]^T$, $H_0 = \mathrm{diag}(0.02, 0.02, 0.03, 0.01)$, $H_w = 0.1I_2$, and $H_v = \mathrm{diag}(0.05, 0.05, 0.01)$, with at most $q = 15$ generators. For $k = 1, \ldots, 350$, the signals are

$$u_k = [\cos(0.1k), \sin(0.1k)]^T,$$
$$\varrho_k = 0.1[\sin(0.2k), \cos(0.3k)]^T,$$
$$v_k = [0.05\cos(0.2k), 0.05\sin(0.3k), 0.01\sin(0.13k)]^T.$$

The estimated pitch is limited to $[-\pi/6, \pi/6]$; $u_0 = 0$. The state bounds $\bar{x} = [1, 0.2, 5, \pi/6]^T$ give $H_p \approx \mathrm{diag}(0, 0, 0.00269, 0)$.

To reduce direct noise injection, $N_S = 0.1C^T I_S$ and $T_S = I_4 - N_S I_S C$ are prescribed. The observer LMIs with $[E_i H_w, H_p(:, 3)]$, DH_v, and a common P yield $\gamma_\xi = 1.44967$. For f_k, the same gains give $\gamma_f = 3718.88$; the input coordinates differ. The scaled LMI matrices are negative definite. For every admissible measurement mask, the eigenvalues of the observer error matrices at both model vertices have absolute values below one, with a maximum of 0.999543. Theorem 1 requires correct isolation within each segment with a fixed mask.

Both detectors use the same prediction set, updated only with trusted depth measurements. Its zonotope/interval box intersection gives c^-_k and diagonal H^-_k. An attack is detected when $|y^d_{k,i} - e_i^T C c^-_k| > r_{k,i}$, with $r_{k,i} = \|e_i^T [CH^-_k, DH_v]\|_1$. The T-N-L observer provides state estimates; c^-_k is used only for detection. EDM uses $\beta = 2.5$, $z_k = \mathrm{diag}(\theta_{k,1}, \theta_{k,2}, 1)$, and $\theta_{k,i} = 0.001 + 0.099\,\mathrm{rand}$ with seed 1. Attacks are added to encoded packets before decoding. In the main case, the attacks are $0.1\cos(0.2k)$ on channel 1 for $50 \le k \le 140$ and $0.1\sin(0.2k)$ on channel 2 for $160 \le k \le 240$. The intermittent case uses $0.08\cos(0.2k)$ on channel 1 for $80 \le k \le 110$ and $160 \le k \le 180$.

The simulation results are presented in Figs. 3–6. Fig. 3 compares the estimates of heave velocity, pitch velocity, depth, and pitch angle. During attacks, the adaptive observer without EDM exhibits larger velocity and pitch angle deviations. The proposed method reduces these deviations and keeps estimates closer to the true states

TABLE I
Detected/attacked samples. Both detectors have zero false alarms in these runs.

Case/channel	With EDM	Without EDM
Main, channel 1	90/91 (98.90%)	24/91 (26.37%)
Main, channel 2	81/81 (100.00%)	14/81 (17.28%)
Intermittent, channel 1	52/52 (100.00%)	1/52 (1.92%)

TABLE II
RMS estimation error J over 350 samples.

Configuration	Main	Intermittent
EDM + adaptive masking	0.0477	0.0541
No EDM + adaptive masking	0.0696	0.0637
EDM, no masking	91.7305	24.7783
No EDM, no masking	0.0841	0.0642
Observer without attacks	0.0501	0.0501

despite measurement noise, even when adaptive attack isolation is already employed.

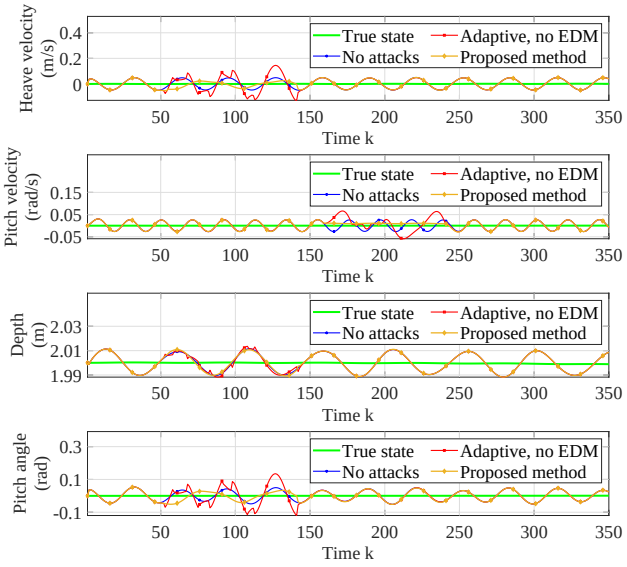


Fig. 3. State estimation results under sparse sensor attacks.

Fig. 4 compares detection without EDM (left) and with EDM (right). A flag of 1 denotes detection, and 0 denotes no detection. Without EDM, many attacked samples remain undetected. With EDM, detection covers almost the entire attack intervals. The detection rates on both channels increase from 26.37% and 17.28% to 98.90% and 100.00%, respectively (Table I). Both detectors have zero false alarms outside the attack intervals in these runs.

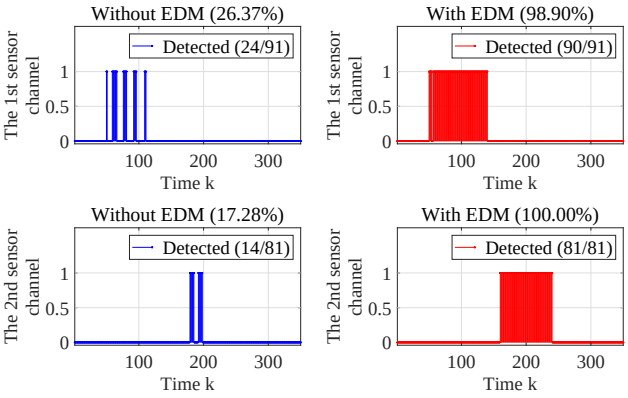


Fig. 4. Attack detection results with and without EDM.

Equation (12) scales packet attacks by $G_k = e^{\beta_k \|u_{k-1}\|_2^2} z_k^{-1}$, potentially large for $\beta = 2.5$ and the small $\theta_{k,i}$. The "EDM, no masking" ablation disables isolation, retaining amplified measurements in the observer update. Its large RMS values (Table II) and errors (Fig. 5, upper) do not demonstrate instability of EDM or describe the complete proposed method, which rejects detected channels through zonotopic tests and adaptive masking. In the lower panel, adaptive attack isolation is enabled for both observers. Adding EDM further reduces the error through improved attack detection. Table II reports $J = \sqrt{350^{-1} \sum_k \|x_k - \hat{x}_k\|_2^2}$. The complete proposed method gives 0.0477 (main) and 0.0541 (intermittent).

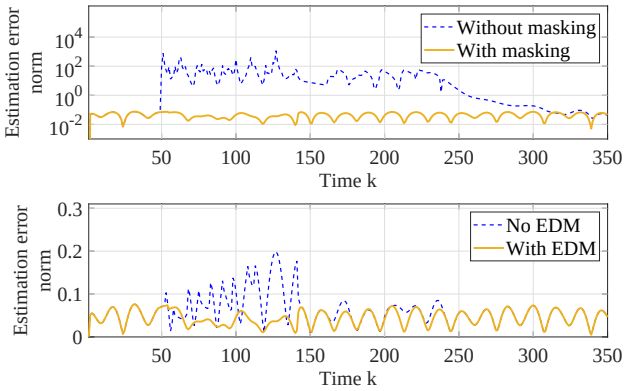


Fig. 5. Estimation error comparison for EDM and adaptive masking.

Fig. 6 tests two smaller, separated attacks on channel 1. The raw attack remains below $2r_{k,1}$ throughout both intervals. EDM detects all 52 attacked samples, whereas the detector without EDM detects only one, demonstrating the advantage of EDM for these smaller attacks. Ten additional realizations per case give mean error reductions of 34.77% and 18.11% for the main and intermittent cases, respectively, relative to adaptive estimation without EDM. These results hold within the prescribed uncertainty bounds. They do not establish robustness to missed amplified attacks or stability under arbitrary mask switching.

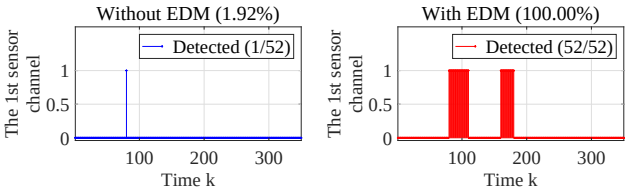


Fig. 6. Attack detection results under intermittent attacks.

V. Conclusion

This paper has proposed an adaptive T-N-L observer with EDM and zonotopic analysis for secure state estimation of AUVs. The robustness result requires correct isolation within segments with fixed masks. Simulations with trusted depth sensing show improved detection and reduced estimation error. Sensing only velocity does not support detectability of all four states. The proposed framework addresses secure state estimation under sparse sensor attacks and bounded uncertainties. Attacks before encoding, replay, communication delays, and packet loss are beyond the current scope and will be investigated in future work.

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